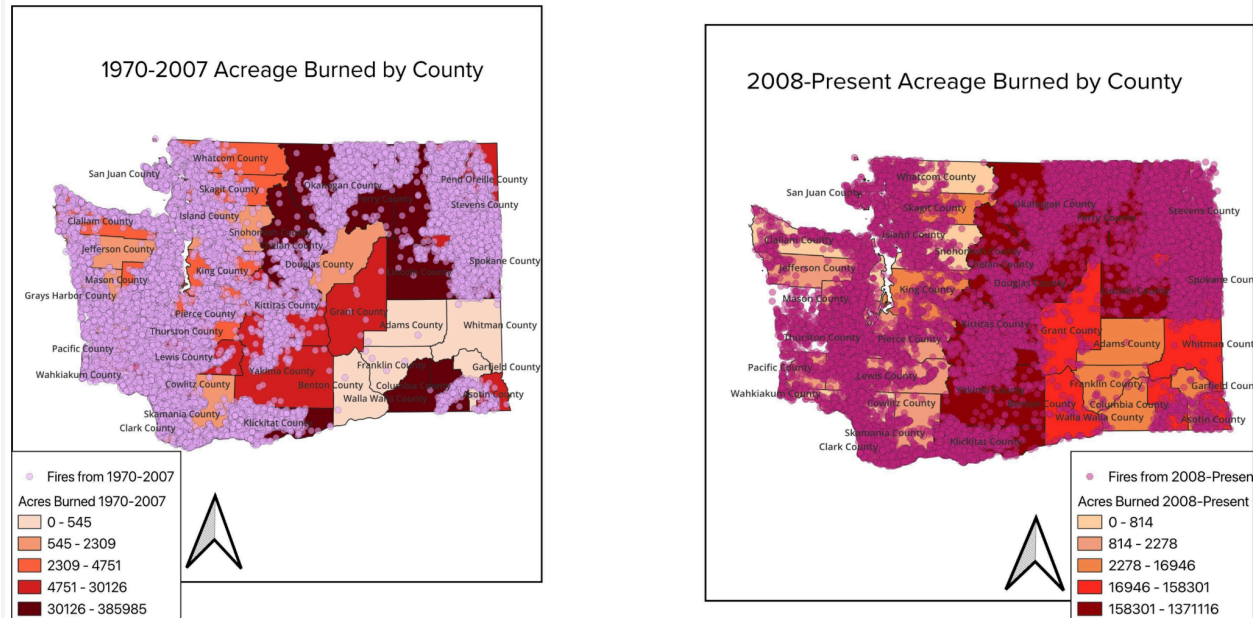


How has the average acreage burned and fire severity on lands protected by the Washington State Department of Natural Resources changed between 1970–2007 and 2008–present?

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Geog 360
EPSG 2927
3/11/26



Introduction

Wildfires are a natural and critical component of forest ecosystems, influencing vegetation patterns, wildlife habitats, and nutrient cycles. However, in recent decades, wildfire frequency, size, and intensity have increased in many of the central regions of Washington, raising concerns about their ecological and economic impact. Understanding how wildfire behavior has changed over time is essential for forest management, risk mitigation, and policy development. This study focuses on lands managed by the Washington State Department of Natural Resources (DNR), which encompasses a mix of forests, grasslands, and other natural areas. Many of the wildfires found in this data are created by humans in order to preserve forest and vegetation across the state. These lands are subject to specific management practices aimed at both ecological health and public safety. The central research question guiding this report is: How has the total acreage burned per county and fire severity on lands protected by the Washington State Department of Natural Resources changed between 1970-2007 and 2008-present. To address this question, the study compares the two time periods using spatial visualizations creating two different maps representing acreage burned and individual fires logged throughout Washington State. By examining these trends, this map aims to identify changes in fire behavior over the last 50 years and provide insights for future DNR management strategies.

Methodology

Fire incident data for DNR-protected lands from 1970 to present were obtained from the Washington State Department of Natural Resources GIS Open Data. The data was then cleaned in Microsoft Excel and added into QGIS. For each fire, relevant attributes were included, such as data, location, acreage burned, severity, and type of fire. For this study, acreage burned was used on the choropleth map. The data was split into two periods, split by the Washington State Department Open Data, with 1970-2007 making up one dataset and 2008-present making up the other. These reflected historical trends and recent changes associated with updated environmental management practices and climate shifts. The data was then joined with the Counties shapefile from Lab 3 and acreage burned was summarized for each present county. Greys Harbor County and Pend Oreille County did not include data from 2008-Present. Two maps were generated for each period, with the same legends producing identical map formats. The dot map represents each fire event and the choropleth contains shaded county regions according to total acreage burned across the state. Average acreage burned per county was calculated for each period, summarizing 38,000 individual fires found in each dataset.

Results

After observing the results, the maximum number of acres burned in an individual county increased by approximately one million acres. The dot map produces a very similar shape in each time period, with a little more density from 2008 to the present, representing an increase in individual fires in Washington State. There is also a pattern of significantly more acreage burned in the center of the state in recent times, with darker regions filling the length of the state in the middle. The southeast corner also shows 2008 to the present rising in the number of acreage burned despite only a small increase in individual fires. The 1970 to 2007 map shows a large number of individual fires in the southwest compared to recent years, but still a low acreage burned. On the other hand, spatial patterns show similar numbers of individual fires, but more widespread impact from 2008 onward, again showing a large increase in severity of recent individual fires. Overall, average acreage burned has increased significantly in recent years, despite not also increasing in individual fires throughout the state. This may be due to new land management strategies, by letting fires burn more and longer before being put out, or due to a lack of precipitation over the more recent years across Washington State.